

POKHARA UNIVERSITY

Level: Bachelor

Semester: Spring

Year : 2024

Programme: BE

Full Marks: 100

Course: Computer Graphics and Multimedia

Pass Marks: 45

Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Explain the construction of CRT monitor and how images are displayed here using Raster scan system. 7
- b) Derive Bresenham Mid Point circle drawing algorithm formula including initial decision parameter. 8
2. a) Define the term shearing in 2D transformation. Reflect a triangle (2,3), (5,3) and (4,4) about the line $y=x$. 7

OR

Derive the transformation matrix for reflection about the line $y=mx+b$.

- b) Why do we need clipping? Explain in detail about Cohen-Sutherland line clipping algorithm. 8
3. a) Describe Octree representation for 3D objects. Differentiate between parallel and perspective projection. 7
- b) Differentiate between ISM and OSM. Explain Scan Line method of back face detection. 8
4. a) What do you mean by surface rendering and shading? Compare and contrast Gouraud and Phong shading. 7
- b) What is MIDI sound? Compare between MIDI and digital audio. 8
5. a) Discuss on BMP, JPEG and PNG file formats. Also, explain the methods of controlling animation. 7

OR

Define video codec and Frame-rate. Explain how video resolution and color depth affect video quality.

- b) Describe the key features and use cases of three different multimedia authoring tools. How do these tools assist in the creation of multimedia content? 8
6. a) What is video? Explain various video compression technology? 7
- b) What are lossy and lossless compression techniques? Explain the JPEG compression techniques. 8

7. Write short notes on: (Any two)

2×5

- a) DDA Vs. Bresenham's line drawing Algorithm.
- b) Composite Transformation
- c) OpenGL and its primitives

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Attempt all the questions.

1. a) What are GPUs? Compare raster scan systems with random scan systems in terms of working principles and applications. 7
- b) Explain color generation technique in CRT monitor for both random and raster. 8
2. a) Show that: (i) Two successive translations are additive, and (ii) Two successive scaling are multiplicative. 8

OR

Derive a composite transformation matrix for reflecting an object about any arbitrary line $y=mx+c$ in 2D.

- b) Explain Cohen Sutherland line clipping algorithm with relevant example with intersection calculation. 7
3. a) Differentiate between parallel and perspective projection. Calculate the co-ordinates of Bezier curve described by 4 control points $P_0(0,0)$, $P_1(1,2)$, $P_2(3,3)$, $P_3(4,0)$ and approximate with 5 lines segments. 7
- b) Differentiate between ISM and OSM. Explain depth buffer method of hidden surface removal. 8
4. a) What is Mach band effect? Explain how Gouraud shading calculates vertex colors and interpolates them across a polygon. 7
- b) Define multimedia and explain its major components. Explain the concept of morphing in animation. 8
5. a) Define video codecs and explain their role in digital video processing. Also discuss the importance of frame rate, resolution and color depth in video quality. 7

OR

Describe the process of audio sampling and quantization. Why are they important in digital audio processing?

- a) What are multimedia authoring tools? List and describe two commonly used authoring tools and their applications. 8
- 6. b) How the image compression is done? Explain it with reference to JPEG format. 8
- c) What are psychoacoustic principles? Describe the key features of MPEG, MPEG-2, MPEG-4, and H.264/AVC video compression standards. 7
- 7. Write short notes on: (Any two) 2×5
 - a) Video Controller
 - b) Window Vs Viewport
 - c) OpenGL

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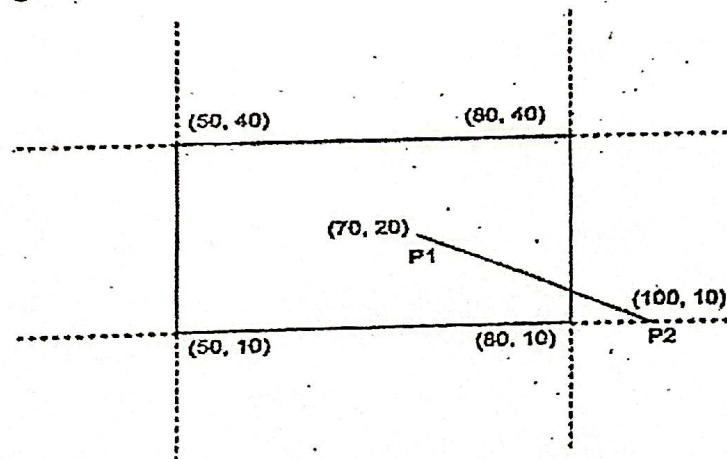
Attempt all the questions.

1. a) What is a GPU and how it assist the CPU to enhance the computer performance? What do you mean by virtual reality and augmented reality? 7
- b) Explain the color generation technique in CRT monitor for both random and raster. 8
2. a) What is a 8-way symmetry and how it helps to reduce the complexity of the circle drawing algorithm. Digitize a circle radius 10 at center (0, 0). 8
- b) Digitize a line having end points (18,10) to (30,18) using BLA algorithm and how this is better than DDA. 7

OR

Derive the equation of DDA line drawing algorithm for all three cases of slope.

3. a) Apply Cohen Sutherland Line Clipping Algorithm to clip the following case. 8



OR

Window port is given by (100, 100, 300, 300) and viewport is given by (50, 50, 150, 150). Convert the window port coordinate (200, 200) to the view port coordinate.

- b) Differentiate between 2D and 3D graphics. Derive the relation for three- dimensional translation and rotation. 7
- 4. a) Explain the different types of Projection with expression and matrix representation. 8
- b) What is the role of ray tracing in visible surface detection? Explain how scan line algorithm is used for back face detection. 7
- 5. a) What do you mean by multimedia? Differentiate between digital audio sound and MIDI. 8
- b) What is animation? Explain the concept of morphing in animation. 7
- 6. a) What is lossy and lossless compression? Explain how JPEG compression is done? 8
- b) What is perceptual audio coding? Explain the video compression standards in briefly. 7
- 7. Write short notes on: (Any two) 2×5
 - a) Grouard Shadding
 - b) Open GL
 - c) Multimedia Workstation

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Attempt all the questions.

1. a) If a pixel is accessed from the frame buffer with an average access time of 300ns then will this rate produce an un-flickering effect for the screen size of 640×480 . 7
- b) Digitize the circle with radius 10. 8
2. a) Rotate the $\triangle ABC$ formed by A (7,15), B (5, 8), C (10, 10) by 90° anticlockwise about (5, 8). 7
- b) Explain Cohen Sutherland line clipping algorithm with relevant examples with intersection calculation. 8

OR

What is window to viewport transformation? Derive a matrix for window to viewport transformation explaining each steps.

3. a) What is projection? Derive a transformation matrix for perspective projection. 8
- b) Differentiate between ISM and OSM. Explain scan line method of hidden surface removal. 7
4. a) What is the drawback of Goraud Shading and how can that be corrected using Phong Shading? 8
- b) What is multimedia system? Explain basic primitive of multimedia system. 7
5. a) What is MIDI? Explain MIDI reception mode in detail. 7
- b) What is compression? Explain each steps of JPEG compression techniques. 8

OR

Explain each steps of AAC compression technique.

6. a) Explain real time requirements of multimedia system. 7

- b) What are different kinds of multimedia authoring tools? 8
7. Write short notes on: (Any two) 2×5
- a) DDA algorithm
 - b) MPEG
 - c) MP3 compression